

ZACHARY GOLD



APPOINTMENTS

2022
|
Present

Ocean Molecular Ecology Group Lead, Research Scientist

NOAA Pacific Marine Environmental Laboratory

📍 NOAA PMEL, Seattle, WA

2022

Molecular Ecology Lead

California Cooperative Oceanic Fisheries Investigations (Scripps Institution of Oceanography & NOAA SWFSC) & Southern California Coastal Watershed Research Project

📍 UCSD SIO, La Jolla, CA

2020
|
2022

Postdoctoral Scholar

NOAA Northwest Fisheries Science Center & University of Washington – The Cooperative Institute for Climate, Ocean, and Ecosystem Studies

📍 NOAA Northwest Fisheries Science Center, Seattle, WA



PROFESSIONAL SERVICE

2025
|
Present

DOC NOAA CRADA with McLane Research Laboratories

Cooperative Research & Development Agreement with McLane Research Laboratories, Inc. for Operationalizing McLane Robotic Cartridge Sampling Instrument (RoCSI) for Extended Mooring Deployments

2024
|
Present

Fish & Invertebrate Scientific Expert Working Group of the Environmental Monitoring Guidance for Offshore Wind

2023
|
Present

Co-Lead NOAA 'Omics Technical Portal

2022
|
Present

CalCOFI Steering Committee

2022
|
Present

NOAA 'Omics Working Group OAR representative

2022
|
Present

Co-Chair NOAA Environmental Biomolecular Observations Subcommittee

2022
|
Present

NOAA Data and Bioinformatics Subcommittee OAR representative

2022
|
Present

Visiting Scientist, Scripps Institution of Oceanography

CONTACT

✉ zachary.gold@noaa.gov

🌐 <https://github.com>

[/marinednadude](#)

🔗 <https://www.pmel.noaa.gov>

[/ocean-molecular-ecology/](#)

🔗 <https://zackgold.org/>

in <https://www.linkedin.com/in>

[/zack-gold-a3035091](#)

Last updated on February 23, 2026.

2020
|
Present

NOAA Metagenomics and Computing Subcommittee

2020
|
Present

California Molecular Methods Working Group



PROFESSIONAL AFFILIATIONS

2025
|
Present

Member, Marine Technology Society eDNA Technology Committee

2021
|
Present

Member, Assoc. for the Sciences of Limnology & Oceanography

2019
|
Present

Research Associate, Natural History Museum of LA County



EDUCATION

2015
|
2020

PhD.

PhD, Ecology and Evolutionary Biology.

📍 UCLA

- Thesis – Design and Implementation of Environmental DNA Metabarcoding Methods for Monitoring the Southern California Marine Protected Area Network

2011
|
2015

B.S.

Bachelor of Science in Marine Science, Honors. Minor in Environmental Systems Engineering.

📍 Stanford University

- Thesis – Long Term Growth Rates and Effects of Bleaching in *Acropora hyacinthus*



AWARDS AND HONORS

2024

Metabarcoding and Metagenomics Editor's Choice Award

2020

Los Angeles Sustainability Collaborative Fellowship

2019

Heal the Bay Super Healer Award

2019

NSF Graduate Research Internship Project Award

2016

National Geographic Young Explorer Program

2015 **National Science Foundation Graduate Research Fellowship**

2015 **UCLA Graduate Dean's Scholar Award**

2014 **Moris K. Udall and Stewart L. Udall Scholarship**

2013 **NOAA Ernest F. Hollings Undergraduate Scholarship**

2012 **Marine Technology Society Undergraduate Scholarship**



GRANTS

2025
|
2027 **Unlocking archived DNA to investigate climate impacts on zooplankton communities in the Southern California Current. SCCOOS (\$690K).**
Dec. 2025 – Dec. 2027.

2025 **eDNA assessments of marine biodiversity to evaluate the effectiveness of marine protected areas. La Kretz Center for California Conservation Science (\$1,400).**
Dec. 2015 – Dec. 2016.

2024
|
2028 **Unlocking archived DNA to investigate climate impacts on zooplankton communities in the Southern California Current. SCCOOS (\$100K).**
August 2024 – Dec. 2028.

2024
|
2028 **Integrated Arctic Ecosystem Toolkit to Enhance Forecasting for Climate Resilience. NOAA GOMO Arctic Research Program (\$276K).**
August 2024 – Dec. 2028.

2024
|
2026 **Development of gene expression indicators of ocean acidification stress in pteropods (*Limacina helicina*) in the CA Current. NOAA OAP (\$109K).**
April 2024 – Dec. 2026.

2024
|
2025 **Developing novel eDNA qPCR assay for rapid assessment of sunflower monitoring and recovery efforts. Association of Zoos and Aquariums (\$20K).**
Oct. 2024 – Oct. 2025.

2024 **The Individual and Synergistic Effects of Four Anthropogenic Stressors on Coral Reef Ecosystem Health. NSF Graduate Research Fellowship Program (\$138,000)**
Sept. 2015 – Dec. 2020.

2023
|
Present

Using 'omics to monitor effects of rapidly changing oceans on West Coast, Alaska, and Arctic marine ecosystems NOAA 'Omics Program implemented through PMEL base funding (\$1.94M).
FY23-25, and ongoing (pending Congressional appropriations).

2022
|
2027

MMARINeDNA: Marine Mammal Remote detection via INnovative environmental DNA sampling. Program of the University Research Initiative (MURI) (\$7,500,000)
March 2022 – March 2027. Department of Defense Multidisciplinary Research. PIs are Ryan Kelly and Brice Semmens, I am grant co-author.

2020

Investigating Patterns of Larval Fish Community Dynamics Over the Past Two Decades Using a Novel Application of Environmental DNA. La Kretz Center for California Conservation Science (\$4,800)
Jan. 2020 – Dec. 2020.

2018
|
2019

Investigating Patterns of Larval Fish Community Dynamics Over the Past Two Decades Using a Novel Application of Environmental DNA. National Science Foundation GRIP Award (\$5000)
April 2018 – April 2019.

2018
|
2019

Citizen Scientist's Message in a Bottle: Rapid eDNA assessments of marine biodiversity to evaluate the effectiveness of Los Angeles' marine protected areas. UCLA EEB Department Grant (\$3,000)
April 2018 – April 2019.

2018
|
2019

Rapid eDNA assessments of marine biodiversity to evaluate the effectiveness of marine protected areas. Resource Legacy Fund – Marine Conservation Grant (\$15,000)
Jan. 2018 – Dec. 2019.

2017
|
2018

Rapid eDNA assessments of marine biodiversity to evaluate the effectiveness of marine protected areas. UCLA EEB Department Grant (\$1,500)
April 2017 – April 2018.

2016
|
2017

Citizen Scientist's Message in a Bottle: Rapid eDNA assessments of marine biodiversity to evaluate the effectiveness of Los Angeles' marine protected areas. La Kretz Center for California Conservation Science (\$2,400)
Dec. 2016 – Dec. 2017

2016
|
2017

Rapid eDNA assessments of marine biodiversity to evaluate the effectiveness of marine protected areas. National Geographic Society – Young Explorer Grant (\$4,800)
Sept. 2016 – Sept. 2017.

2016
|
2017

Rapid eDNA assessments of marine biodiversity to evaluate the effectiveness of marine protected areas. Sigma Xi Grants-in-Aid (\$900)
April 2016 – April 2017.

2016
|
2017

eDNA assessments of marine biodiversity to evaluate the effectiveness of marine protected areas in the California Channel Islands.. International SeaKeeper Society Support (\$2,000 in vessel time)

April 2016 – Dec. 2017.

2015
|
2020

Rapid eDNA assessments of marine biodiversity to evaluate the effectiveness of marine protected areas in the California Channel Islands. UC Catalyst Grant for Conservation Genomics Grant (\$43,000)

Mar. 2016 – Mar. 2017. PI was Paul Barber, I am grant co-author.

2015
|
2016

Rapid eDNA assessments of marine biodiversity to evaluate the effectiveness of marine protected areas. UCLA EEB Department Grant (\$1,500)

April 2016 – April 2017

PEER REVIEWED PUBLICATIONS

Preprint

Guidance and best practices for species identification using eDNA metabarcoding-When do you call a cod a cod?
Watts, A., Gold, Z., Patin, N., Adams, N., Baker, J., El Baidouri, F., Grey, E., Holmes, A., Jacobson, K., Jungbluth, S. and Kajita, T.,
 Metabarcoding and Metagenomics

Links: [DOI](#)

2025

The essential role of large research vessels in marine ecosystem observations and ocean sustainability.
Satterthwaite, E.V., Field, J.C., Fassbender, A.J., Aceves-Medina, G., Bograd, S.J., Hazen, E.L., Patin, N.V., Gold, Z., Barbeau, K.A., Kinkade, D. and Shepherd, A.
 Limnology and Oceanography

Links: [DOI](#)

2025

Using eDNA to Supplement Population Genetic Analyses for Cryptic Marine Species: Identifying Population Boundaries for Alaska Harbour Porpoises.
Parsons, K.M., May, S.A., Gold, Z., Dahlheim, M., Gabriele, C., Straley, J.M., Moran, J.R., Goetz, K., Zerbini, A.N., Park, L. and Morin, P.A
 Molecular Ecology

Links: [DOI](#)

2025

What is eDNA method standardisation and why do we need it?
Theroux, S., Sepulveda, A., Abbott, C.L., Gold, Z., Watts, A.W., Hunter, M.E., Klymus, K.E., Hirsch, S.L., Craine, J.M., Jones, D.N. and Brown, R.J.
 Metabarcoding and Metagenomics

Links: [DOI](#)

2024 **The MIEM guidelines: Minimum information for reporting of environmental metabarcoding data.**
Klymus, K.E., Baker, J.D., Abbott, C.L., Brown, R.J., Craine, J.M., Gold, Z., Hunter, M.E., Johnson, M.D., Jones, D.N., Jungbluth, M.J. and Jungbluth, S.P.
📍 Metabarcoding and Metagenomics

Links: [DOI](#)

2024 **Archived DNA reveals impacts of a marine heatwave on fish assemblages over multiple decades.**
Gold, Z., Kelly, R., Shelton, A.O., Goodwin, K., Gallego, R., Parsons, K., Thompson, L., Kacev, D., Thompson, A., Barber, P.H.
📍 Environmental DNA

Links: [DOI](#)

2024 **rCRUX: A rapid and versatile tool for generating metabarcoding reference libraries in R.**
Curd, E.E., Gal, L., Gallego, R., Silliman, K., Nielsen, S. and Gold, Z.
📍 Environmental DNA

Links: [DOI](#)

2024 **Hidden in plain sight: the invasive macroalga *Caulerpa prolifera* evades detection by environmental DNA methods due to negligible shedding.**
Waters, T., Langlois, K., Gold, Z., Theroux, S., & Eagle, R.
📍 Environmental DNA

Links: [DOI](#)

2024 **A pile of pipelines: An overview of the bioinformatics software for metabarcoding data analyses.**
Hakimzadeh, A., Abdala Asbun, A., Albanese, D., Bernard, M., Buchner, D., Callahan, B., Caporaso, J. G., Curd, E., Djemiel, C., Brandström Durling, M., Elbrecht, V., Gold, Z., Gweon, H. S., Hajibabaei, M., Hildebrand, F., Mikryukov, V., Normandeau, E., Özkurt, E., M. Palmer, J. ... Anslan, S.
📍 Molecular Ecology Resources

Links: [DOI](#)

2024 **Toward a national eDNA strategy for the United States.**
Kelly, R. P., Lodge, D. M., Lee, K. N., Theroux, S., Sepulveda, A. J., Scholin, C. A., Craine, J. M., Andruszkiewicz Allan, E., Nichols, K. M., Parsons, K. M., Goodwin, K. D., Gold, Z., Chavez, F. P., Noble, R. T., Abbott, C. L., Baerwald, M. R., Naaum, A. M., Thielen, P. M., Simons, A. L. ... Weisberg, S. B.
📍 Environmental DNA

Links: [DOI](#)

2024 **Environmental DNA metabarcoding reveals distinct fish assemblages supported by seagrass (*Zostera marina* and *Zostera pacifica*) beds in different geographic settings in Southern California.**
Waters, T., Gold, Z., Obaza, A., Ambrose, R. F., & Eagle, R. A.
📍 PLOS One

Links: [DOI](#)

2023 **Signal and noise in metabarcoding data.**
Gold, Z., Shelton, A.O., Casendino, H.R., Duprey, J., Gallego, R., Van Cise, A., Fisher, M., Jensen, A.J., D'Agnese, E., Andruszkiewicz Allan, E. and Ramón-Laca, A.,
PLOS One

Links: DOI

2023 **Quantifying impacts of an environmental intervention using environmental DNA.**
Andruszkiewicz Allan, E., Kelly, R. P., D'Agnese, E. R., Garber-Yonts, M. N., Shaffer, M. R., Gold, Z., & Shelton, A. O.
Ecological Applications

Links: DOI

2023 **A Comparison of Biomonitoring Methodologies for Surf Zone Fish Communities**
Gold, Z., Koch, M.Q., Schooler, N.K., Emery, K.A, Dugan, J.E., Miller, R.J., Page, H.M, Schroeder, D.M., Hubbard, D.M., Madden, J., Whitaker, S.G., Barber, P.H.
PLOS One

Links: DOI

2023 **Toward identifying the critical ecological habitat of larval fishes: An environmental DNA window into fisheries management. In Frontiers in Ocean Observing: Emerging Technologies for Understanding and Managing a Changing Ocean.**
Satterthwaite, E.V., A.E. Allen, R.H. Lampe, Z. Gold, A.R. Thompson, N. Bowlin, R. Swalethorp, K.D. Goodwin, E.L. Hazen, S.J. Bograd, S.A. Matthews, and B.X. Semmens.
Oceanography

Links: DOI

2022 **Toward quantitative metabarcoding.**
Shelton, A.O., Gold, Z., Allen, E., D'Agnese, E., Van Cise, A., Gallego, R., Parsons, K., Duprey, J., Casedino, H., Garber-Yonts, M., Fisher, M., Kelly, R.
Ecology

Links: DOI

2022 **A manager's guide to using eDNA metabarcoding in marine ecosystems.**
Gold, Z., Wall, A.R., Schweizer, T.M., Pentcheff, N.D., Curd, E.E., Barber, P.H., Meyer, R.S., Wayne, R., Stolzenbach, K., Prickett, K. and Luedy, J.
PeerJ

Links: DOI

2021 **Improving Metabarcoding Taxonomic Assignment: A Case Study of Fishes in a Large Marine Ecosystem.**
Gold, Z., Curd, E., Goodwin, K., Choi, E., Fable, B., Thompson, A., Burton, R., Kacev, D., and Barber, P.
Molecular Ecology Resources

Links: DOI

- 2021 **eDNA metabarcoding as a biomonitoring tool for marine protected areas**
Gold, Z., Sprague, J., Kushner, D. J., Zerecero Marin, E., & Barber, P. H.  PLOS One
[Links: DOI](#)
- 2021 **MiSebastes: An eDNA metabarcoding primer set for rockfishes (genus Sebastes).**
Min, M. A., Barber, P. H., & Gold, Z.  Conservation Genetics Resources
[Links: DOI](#)
- 2021 **Short-lived detection of an introduced vertebrate eDNA signal in a nearshore rocky reef environment.**
Ely, T., Barber, P. H., Man, L., & Gold, Z.  PLOS One
[Links: DOI](#)
- 2021 **eDNA Captures Depth Partitioning in a Kelp Forest Ecosystem.**
Monuki, K., Barber, P., & Gold, Z.  PLOS One
[Links: DOI](#)
- 2021 **Environmental DNA in a Global Biodiversity Hotspot: Lessons from Coral Reef Fish Diversity Across the Indonesian Archipelago.**
Marwayana, O.N., Gold, Z., Meyer C., and Barber, P.H.  Environmental DNA
[Links: DOI](#)
- 2021 **Characterizing Industrial and Artisanal Fishing Vessel Catch Composition Using Environmental DNA and Satellite-Based Tracking Data.**
Willette, D. A., Navarrete-Forero, G., Gold, Z., Lizano, A. M. D., Gonzalez-Smith, L., & Sotil, G.  Foods
[Links: DOI](#)
- 2021 **The CALeDNA program: Citizen scientists and researchers inventory California's biodiversity.**
Meyer R., Ramos M., Lin M., Schweizer T., Gold Z., Ramos D., Shirazi S., Kandlikar G., Kwan W., Curd E., Freise A., Parker J., Sexton J., Wetzer R., Pentcheff N., Wall A., Pipes L., Garcia-Vedrenne A., Mejia M., Moore T., Orland C., Ballare K., Worth A., Beraut E., Aronson E., Nielsen R., Lewin H., Barber P., Wall J., Kraft N., Shapiro B., Wayne R.  California Agriculture
[Links: DOI](#)
- 2020 **Landscape analyses using eDNA metabarcoding and Earth observation predict community biodiversity in California.**
Lin, M., Simons, A.L., Curd, E.E., Harrigan, R.J., Schneider, F.D., Ruiz-Ramos, D.V., Gold, Z., Osborne, M.G., Shirazi, S., Schweizer, T.M. and Moore, T.N.  Ecological Applications
[Links: DOI](#)

2020 **eDNA metabarcoding bioassessment of endangered fairy shrimp (*Branchinecta* spp.).**
Gold, Z., Wall, A.R., Curd, E.E., Kelly, R.P., Pentcheff, N.D., Ripma, L., Barber, P.H. and Wetzler, R.

📍 Conservation Genetics Resources

Links: [DOI](#)

2020 **Design and implementation of environmental DNA metabarcoding methods for monitoring the Southern California marine protected area network**

Gold, Z.

📍 Doctoral Dissertation

Links: [Link](#)

2019 **Anacapa Toolkit: an environmental DNA toolkit for processing multilocus metabarcode datasets.**

Curd, E. E., Gold, Z., Kandlikar, G. S., Gomer, J., Ogden, M., O'Connell, T., Pipes, L., Schweizer, T. Rabichow, L., Lin, M., Shi, B., Barber, P.H., Kraft, N., Wayne, R., Meyer, R.

📍 Methods in Ecology and Evolution

Links: [DOI](#)

2018 **Long-term growth rates and effects of bleaching in *Acropora hyacinthus*.**

Gold, Z., & Palumbi, S. R.

📍 Coral Reefs

Links: [DOI](#)

2018 **ranacapa: An R package and Shiny web app to explore environmental DNA data with exploratory statistics and interactive visualizations.**

Kandlikar, G. S., Gold, Z. J., Cowen, M. C., Meyer, R. S., Freise, A. C., Kraft, N. J., Moberg-Parker, J., Sprague, J., Kushner, D. J., Curd, Emily E

Links: [DOI](#)



TECHNICAL AND GOVERNMENT REPORTS

2025 **Introduction to Developing DNA Reference Barcode Sequences**

Gold, Z., Brown, S., Collins, A., Girard, M., Goodwin, K., McAllister, S., Meyer, C., Parsons, K., Patin, N., Rouse, G., Sala, L., Satterthwaite, E., Seid, C., Theroux, S., & Wetzler, R.

📍 The West Coast Ocean Biomolecular Observing Network

Links: [DOI](#)

2024 **National Aquatic Environmental DNA Strategy**

Goodwin, K. D., C. M. Aiello, M. Weise, M. Edmondson, K. Fillingham, D. Allen, A. Amerson, M.L. Barton, A. Benson, G. Canonico, G., Z. Gold, J. Gumm, M. Hunter, N. Joffe, R. Lance, A. Larkin, R. Letelier, C. Lipsky, D. McCoskey, C. Morrison, K. Clark, J.A. Darling, A-J. Demery, M. Everett, C. Fletcher-Hoppe, K.M. Nichols, K.M. Parsons, J. Price, K. Puglise, K. Scholl, M.K. Schwartz, A. Sepulveda, J. Shannon, W. Turner, T. White

📍 White House Office of Science, Technology, and Policy

Links: [Link](#)

PRESENTATIONS

- 2025 **Genetic approaches to support the management & conservation of cetaceans.**
Marine Mammal Ecology, University of Alaska Anchorage, AK. Invited Speaker. October 2025. Remote.
- 2025 **Autonomous eDNA Sampler Panel**
NOAA 'Omics 4th In Situ & Autonomous Sampler Technology Workshop. April 2025. Co-Host. Remote
- 2025 **Improving Trust and Reliability in 'Omics and eDNA Methods through FAIR Taxonomic Classification**
NOAA Enterprise Data Management Workshop. May 2025. Remote
- 2025 **Unlocking archived DNA to investigate climate impacts on zooplankton communities in the CA Current**
Ocean Carbon & Biogeochemistry Conference. Invited Speaker. June 2025. Remote.
- 2025 **What goes on the Y axis? 'Omics Measurements to Characterize Biological Responses to OA**
Ocean Acidification + eDNA Workshop hosted by CA OPC & SCCWRP. Invited Speaker. March 2025.
- 2025 **Leveraging eDNA to Survey Large Marine Ecosystems in the Northeast Pacific.**
San Francisco State University, CA. Invited Speaker. March 2025. Remote
- 2025 **qPCR Development Introduction to Unikseq + Geneious Workflow**
California Molecular Methods Working Group. Invited Speaker. May 2025.
- 2025 **Autonomous eDNA sampling to track seasonal hypoxia impacts in Olympic Coast National Marine Sanctuary**
Olympic Coast Intergovernmental Policy Council for the Olympic Coast National Marine Sanctuary. Invited Speaker. January 2025. Remote.
- 2024 **qPCR Development Introduction to Unikseq + Geneious Workflow**
Alaska eDNA Working Group. Invited Speaker. December 2024.
- 2024 **Genetic approaches to support the management & conservation of cetaceans.**
Marine Mammal Ecology, University of Alaska Anchorage, AK. Invited Speaker. October 2024. Remote.
- 2024 **West Coast OBON HABs Monitoring**
NHABON Advanced Methods in HABs Monitoring Seminar Series. Invited Speaker. June 2024. Remote
- 2024 **Operationalizing Metabarcoding Best Practices & Standardization: A NOAA 'Omics Perspective. Invited Speaker.**
3rd National Workshop on Marine eDNA. Invited Speaker. June 2024.

- 2024 **Archived DNA Reveals Marine Heatwave-associated Shifts in Fish and Zooplankton Assemblages.**
Ocean Observing in California Conference, San Diego, CA. May 2024
- 2024 **Mobilizing eDNA for Management in the Northeast Pacific Workshop. Co-host.**
Co-host of webinar. April 2024. Remote.
[Links: Recording](#)
- 2024 **Mobilizing eDNA for Management in the Northeast Pacific Webinar. eDNA 101.**
Co-host of webinar. April 2024. Remote.
- 2024 **eDNA and biodiversity standardization efforts**
UCLA Graduate eDNA Seminar Series. Invited Speaker. March 2024. Remote.
- 2024 **Expanding the Ocean Biomolecular Observing Network to New Locations, Technology, and Data Platforms**
Ocean Sciences Meeting, New Orleans LA. Session Co-Chair. February 2024.
- 2024 **Frontiers in Ocean Biodiversity Observation Through DNA Metabarcoding.**
National Academies: Decadal Survey of Ocean Sciences - Panel on Ocean Life, Gulfport, MS. Invited Speaker. February 2024.
- 2024 **Ocean Acidification Research Success Stories along the North American coast. Invited Speaker.**
GOA-ON North America, Neward, DE. Invited Speaker. January 2024. Remote
- 2023 **Archived DNA reveals marine heatwave-associated shifts in fish and zooplankton assemblages**
PICES, Seattle, WA, USA. October 2023.
- 2023 **eDNA and biodiversity standardization efforts**
PICES Workshop Towards climate-informed ecosystem-based fisheries management by building international collaborations and standardizing indicators, Seattle, WA, USA. Invited Speaker. October 2023.
- 2023 **Operationalizing eDNA for Routine Marine Biomonitoring**
San Francisco State University, San Francisco, CA, USA. Invited Speaker. September 2023.
- 2023 **Operationalizing eDNA for Routine Marine Biomonitoring**
MBARI Seminar, Moss Landing, CA, USA. July 2023.
- 2023 **PMEL Ocean Molecular Ecology**
PNNL Biosciences Seminar, Richland, WA, USA. Invited Speaker. July 2023
- 2023 **Metabarcoding Taxonomic Assignment**
NOAA Metagenomics & Computing Group Meeting. Invited Speaker. June 2023. Remote.
- 2023 **National eDNA Strategy Panel**
Capitol Hill Oceans Week, Washington D.C., USA. June 2023.

- 2023 **eDNA: Leveraging 'Omics To Characterize Ocean Acidification Impacts**
NOAA Ocean Acidification Working Group. May 2023.Remote.
- 2023 **Early Career Federal Ocean Professional Panelist**
OAR Senior Research Council, NOAA NCEP, MD, USA. August 2023.
- 2023 **eDNA: Revolutionizing Ecosystem Monitoring**
UN Decade Dialogue Using eDNA for monitoring and Stewardship in the NE Pacific. May 2023. Remote.
- 2023 **Innovating Ocean Observations:Scaling Sensors to Track Rapid Marine Change**
OceanX Meeting, Silver Spring, MD, USA. Invited Speaker. March 2023.
- 2023 **Making eDNA Count - Towards Quantitative Metabarcoding**
NOAA 'Omics Seminar. Invited Speaker. February 2023. Remote.
[Links: Recording](#)
- 2023 **'Omics Group Overview**
EDD Science Fusion Seminar, NOAA PMEL, Seattle, WA, USA. Invited Speaker. February 2023.
- 2023 **Ecology of eDNA in Marine Systems**
Electric Power Research Institute, Palo Alto, CA, USA. Invited Speaker. February 2023. Remote
- 2023 **West Coast Ocean Biomolecular Observing Network: Integrating and harmonizing molecular observations in support of sustainable marine management**
West Coast Ocean Biomolecular Observing Network Workshop, La Jolla, CA, USA (Hosted by CalCOFI, led workshop). December 2023.
- 2023 **West Coast Ocean Biomolecular Observing Network: Integrating and harmonizing molecular observations in support of sustainable marine management**
CalCOFI Conference, La Jolla, CA, USA (Hosted by CalCOFI, assisted with hosting efforts). December 2023.
- 2022 **Stumbling Head First into the eDNA Revolution: Changing the Paradigm of Ecosystem Monitoring**
Fisheries Genetics Guest Lecture, University Alaska Fairbanks, AK, USA. November 2022. Remote.
- 2022 **Marine Heatwave-Driven Impacts on Fish Assemblages Uncovered Through Archived DNA**
Scripps Institution of Oceanography Ecology Seminar Series, La Jolla, CA, USA. Invited Talk. September, 2022.
- 2022 **Bioinformatics 101: An introduction to bioinformatics for environmental DNA**
2nd National Workshop on Marine Environmental DNA, Costa Mesa, CA, USA. Co-Host. September 2022.

- 2022 **Archived DNA Reveals Marine Heatwave-Associated Shifts in Fish Assemblages**
45th Annual Larval Fish Conference & 13th International Larval Biology Symposium, La Jolla, CA, USA. Co-Host. August 2022.
- 2022 **Making eDNA Count - Towards Quantitative Metabarcoding**
California Molecular Methods Working Group, Costa Mesa, CA, USA. August 2022.
- 2022 **Who's there? & How much? Challenges for environmental DNA.**
Maine-eDNA EPSCoR Annual Symposium, Bigelow Laboratory for Ocean Science, Boothbay, ME, USA. Invited Keynote Speaker. June 2022. Remote.
- 2022 **Linking Salmon Run Timing and endangered Cook Inlet beluga foraging patterns across Cook Inlet through eDNA**
Alaska Chapter American Fisheries Society, Fairbanks, AK, USA. March 2022. Remote.
- 2022 **Revealing endangered Cook Inlet beluga foraging patterns through eDNA**
Alaska Marine Science Symposium, AK, USA. January 2022.
- 2021 **Marine-Heat-Wave-Driven Shifts in Larval Fish Community Dynamics Over the Past Two Decades Revealed Through Metabarcoding**
MBARI, Moss Landing, CA, USA. Invited Speaker. October 2021. Remote.
- 2021 **Combining eDNA And Visual Survey Methods For Comprehensive Biomonitoring Of The Southern California Marine Protected Area Network**
ASLO Remote. Aquatic Science Meeting, Majorca, Spain. June 2021. Remote.
- 2021 **The Anacapa Toolkit**
Genetics and Evolution, NOAA NWFSC, Seattle, Washington. Invited Speaker. March 2021.
- 2021 **Marine-Heat-Wave-Driven Shifts in Larval Fish Community Dynamics Over the Past Two Decades Revealed Through Metabarcoding**
Seattle-Kobe International Academic Exchange Initiative, Kobe, Japan. March 2021. Remote.
- 2021 **Improving Metabarcoding Taxonomic Assignment – A Case Study of Fishes in the California Current Large Marine Ecosystem**
NOAA Bioinformatics Supergroup, NOAA AOML, Miami, Florida. February 2021. Invited Speaker. Remote.
- 2021 **eDNA Metabarcoding - Advances for California Current Large Marine Ecosystem Biomonitoring**
California Molecular Methods Working Group, Costa Mesa, California. February 2021. Invited Speaker. Remote.

- 2020 **eDNA Metabarcoding - Advances for California Current Large Marine Ecosystem Biomonitoring**
Stanford Hopkins Marine Station, Monterey, California. October 2020. Invited Speaker. Remote.
- 2020 **The Anacapa Toolkit**
NOAA Bioinformatics Supergroup, NOAA AOML, Miami, Florida. Invited Speaker. September 2020. Remote.
- 2020 **eDNA Metabarcoding - Advances for California Current Large Marine Ecosystem Biomonitoring**
Conservation Biology Division, NOAA NWFSC, Seattle, Washington. July 2020. Invited Speaker. Remote.
- 2020 **Investigating Patterns of Larval Fish Community Dynamics Over the Past Two Decades**
Ocean Sciences Meeting, San Diego, California. February 2020.
- 2019 **Investigating Patterns of Larval Fish Community Dynamics Over the Past Two Decades**
CalCOFI Conference, La Jolla, California. December 2019.
- 2019 **eDNA: Novel Molecular Techniques for Monitoring MPAs**
Western Society of Naturalists, Ensenada, Baja California, Mexico. November 2019.
- 2019 **eDNA: Novel Molecular Techniques for Monitoring MPAs**
Ocean Protection Council, Sacramento, California. Invited Speaker. June 2019.
- 2019 **Applying Marine eDNA**
eDNA and Global Change Workshop, University of Washington, WA. May 2019.
- 2019 **eDNA: Preliminary Comparison of Methods in Ports of LA and LB**
Natural History Museum of LA County, Los Angeles, California. May 2019.
- 2019 **eDNA: Novel Molecular Techniques for Monitoring Biodiversity**
La Kretz Conservation Genomics Workshop, Stunt Ranch Reserve, CA. March 2019.
- 2019 **Fisheries in the Blue Economy**
Waitt Foundation Blue Economy SIDs Workshop, UCLA, California. February 2019.
- 2019 **eDNA in Vernal Pools: Monitoring Endangered Species and Threatened Aquatic Ecosystems**
UCLA Conservation Genomics Consortium Meeting, UCLA, California. February 2019.
- 2018 **eDNA: Novel Molecular Techniques for Monitoring Aquatic and Marine Ecosystems**
UCGCC eDNA Conference, Asilomar, California. November 2018.
- 2018 **Anacapa Toolkit**
UCLA EEB Graduate Seminar Series, UCLA, California. October 2018.

- 2018 **eDNA: Novel Molecular Techniques for Monitoring MPAs**
MPA Technology Workshop, La Jolla, California. May 2018.
- 2018 **eDNA: Novel Molecular Techniques for Monitoring Biodiversity**
La Kretz Conservation Genomics Workshop, Stunt Ranch Reserve, CA. March 2018.
- 2018 **eDNA: Novel Molecular Techniques for Monitoring the Oceans**
Leonardo Dicaprio Foundation, Los Angeles, CA. February 2018.
- 2018 **eDNA Assessment of Biodiversity: Capabilities and Caveats of Novel Metabarcoding Approaches**
La Kretz Conservation Agency Workshop, Stunt Ranch Reserve. February 2018.
- 2018 **Long Term Growth Rates and Effects of Bleaching in *Acropora hyacinthus***
UCLA Ecology and Evolutionary Biology Graduate Seminar Series. January 2018.
- 2017 **Spatial and Temporal Variation of Environmental DNA in Kelp Forest Ecosystems,**
UCGCC eDNA Workshop, UCLA, California. December 2017.
- 2017 **Developing and Implementing an Aquatic eDNA Methodology**
UCGCC Conference, Blue Oak Reserve, California. August 2017.
- 2017 **Spatial Variation of Environmental DNA in Coral Reef Ecosystems**
Indo-Pacific Fish Conference, Papeete, Tahiti, French Polynesia. October 2017.
- 2017 **Spatial and Temporal Variation of Environmental DNA in Kelp Forest Ecosystems**
Gordon Research Conference Marine Molecular Ecology, HKUST, HK. July 2017.
- 2017 **Spatial and Temporal Variation of eDNA in Kelp Forest Ecosystems**
UCLA EEB Department Graduate Student Seminar Series, UCLA, CA. June 2017.
- 2016 **Environmental DNA: History, Background, Applications, and Bioinformatics**
La Kretz Conservation Genomics Workshop, Stunt Ranch Reserve, CA. March 2017.
- 2015 **Designing and Implementing an Environmental DNA Methodology**
UCLA EEB Department Graduate Student Seminar Series, UCLA, CA. December 2016.
- 2014 **Growth Patterns and Life History Strategies of Thermally Resilient and Susceptible *Acropora hyacinthus* species**
Stanford Honors Thesis Symposium 2015, Stanford University, CA. May 2015.

2013 **Forecasting Ocean Acidification and Pteropod Calcification off the Washington and Oregon Coast**
NOAA Hollings Symposium, NOAA Headquarters, Silver Spring, MD. August 2014.

2013 **Coral Recruitment in Moreton Bay**
Targeted Research Project Symposium, U. Queensland, Brisbane AUS. November 2013.



OUTREACH

2025 **NOAA Science Camp**

2024 **NOAA Science Camp**

2024 **Capital Hill Oceans Week**

2023 **Capital Hill Oceans Week**

2023 **eDNA Community Science with Ocean Research College Academy Program (ORCA)**

2019 **eDNA Community Science with Heal the Bay's MPAWatch Program, eDNA LA River Workshop**

2019 **Heal the Bay MPAWatch Spring Training**

2019 **Heal the Bay MPAWatch Winter Training**

2018 **Heal the Bay MPAWatch Fall Training**

2018 **Photographing Nature: using Photography to Communicate Science and Biodiversity, UCLA EEB Department Graduate Seminar Series**

2014 **Giving up the Bacon: Protecting the Ocean by Going Vegetarian. Conservation Lecture Series**



CONFERENCES AND WORKSHOPS

2026 **West Coast Ocean Biomolecular Observing Network. Remote.**

2026	California Intertidal Biodiversity DNA Barcode Library project Workshop
2025	Ocean Acidification + eDNA Workshop
2025	NOAA 'Omics Workshop: Better Biomolecular Ocean Practices (BeBOP) Protocols & The NOAA 'Omics Technical Portal
2025	NOAA 'Omics 4th In Situ & Autonomous Sampler Technology Workshop. Co-Host. Remote.
2025	Alaska Marine Science Symposium, Anchorage, AK
2025	West Coast Ocean Biomolecular Observing Network. Remote.
2024	Capitol Hill Ocean Week, D
2024	3rd National Marine eDNA Workshop, JHAPL, MD
2024	Ocean Observing in California, San Diego, CA
2024	Ocean Sciences Meeting, New Orleans, LA
2024	National Academies: Decadal Survey of Ocean Sciences, Gulfport, MS
2024	GOA-ON North America, Newark, DE
2023	PICES-2023 Connecting Science and Communities for Sustainable Seas
2023	PICES-2023 Towards climate-informed ecosystem-based fisheries management by building international collaborations and standardizing indicators
2023	Ocean Decade Regional Collaborative Center for the Northeast Pacific Workshop

2023	NOAA OceanX Workshop, DC
2023	Ocean Acidification Program Meeting, La Jolla, CA
2022	West Coast Ocean Biomolecular Observing Network: Integrating and harmonizing molecular observations in support of sustainable marine management
2022	CalCOFI Conference, La Jolla, CA
2022	2nd National Workshop on Marine Environmental DNA, Costa Mesa, CA
2022	45th Annual Larval Fish Conference & 13th International Larval Biology Symposium, La Jolla, CA
2022	Alaska Chapter American Fisheries Society, Remote.
2022	Alaska Marine Science Symposium, Remote.
2021	CalCOFI Conference, Remote.
2021	ASLO Aquatic Science Meeting, Remote.
2021	DNAqua International Conference, Remote.
2020	CalCOFI Conference, Remote.
2020	POGO - Use of eDNA in Marine Environments, Remote.
2020	Ocean Sciences Meeting, San Diego, CA
2019	Western Society of Naturalists, Baja California, Mexico
2019	CALeDNA LA River Workshop, UCLA, CA

2019 **eDNA and Global Change Workshop, University of Washington, WA**

2019 **La Kretz Conservation Genomics Workshop, Stunt Ranch Reserve, CA**

2018 **National Conference on Marine Environmental DNA, Rockefeller Univ. , NY**

2018 **UC Conservation Genomics Consortium eDNA Conference, Asilomar, CA**

2018 **California Marine Protected Areas Technology Workshop, La Jolla, CA**

2018 **La Kretz Conservation Genomics Workshop, Stunt Ranch Reserve, CA**

2018 **La Kretz Conservation Management Workshop, Stunt Ranch Reserve, CA**

2017 **UC Conservation Genomics Consortium eDNA Conference, UCLA**

2017 **Indo-pacific Fish Conference, Papeete, Tahiti, French Polynesia**

2017 **UC Conservation Genomics Consortium Conference, Blue Oak Reserve, CA**

2017 **Gordon Research Conference Marine Molecular Ecology, HKUST, Hong Kong**

2017 **Gordon Research Seminar Marine Molecular Ecology, HKUST, Hong Kong**

2017 **La Kretz Conservation Genomics Workshop, Stunt Ranch Reserve**

2015 **eDNA Working Group Meeting and Workshop, Bangor University, Wales**

+ MENTORSHIP

2026 Present	Hylais Landry, Undergraduate, UW CICOES
2025 Present	Meghan Shea, Postdoctoral Scholar, Stanford
2025 Present	Maddeine Pachenco, Graduate, UCLA
2025 Present	Avery Mathias, Graduate, UW Oceanography
2025	Ryan Moinazad, Undergraduate, USC
2024 Present	Jordana Sevigny, UCSC, PhD. Prof. Alexa Fredstone
2024 Present	Samantha Setta, Postdoctoral Scholar, UW CICOES
2024 Present	Haila Schultz, Graduate, UW Oceanography
2024 Present	Eleanor Crotty, Undergraduate, NOAA Hollings Scholar
2024 2026	Karina Lai, Undergraduate, UW CICOES
2024 2025	Mugdha Chiplunkar, Undergraduate, UW CICOES
2024 2025	Kenna Dailey, Undergraduate, UW CICOES
2023 Present	Dwan Jackson, JSMSU, Masters, Advisor is Prof. Brent Thoma
2023 Present	Mira Abrecht, UCLA, PhD, Advisor is Prof. Dave Jacobs
2023	Laurence Dugal, UWA, PhD, Viva Voce, Defended (2023).
2023	Olivia Kaiaua, Undergraduate, UW CICOES

2022 Present	Maris Goodwin, UA Fairbanks, Masters, Advisor is Prof. Jessica Glass
2022 2023	Tanner Waters, UCLA, PhD – Defended 2023.
2021 2022	Helen Casendino, Undergraduate University of Washington (2021-2022)
2021 2022	Joe Duprey MSc student, University of Washington (2021- 2022)
2021	Rowan Batts, Undergraduate, NOAA Hollings Scholar
2020 2021	Abigail Keller, MSc student, University of Washington (2020- 2021)
2019 2022	Candice Cross, Graduate, UCLA
2019	UCLA Entering Mentoring Training Program Certificate. NIH Supported
2018 2022	Rachel Turba, Graduate, UCLA
2018 2020	Beverly Shih, Undergraduate, UCLA
2018 2020	Nikita Sridhar, Undergraduate, UCLA
2018 2020	Lauren Man, Undergraduate, UCLA
2018 2020	Cristopher Ruano, Undergraduate, UCLA
2018 2019	Abigail Moore, Ph.D student, Universitas Hasanuddin
2017 Present	Erick Zerecero, Graduate, UCLA
2017 Present	Onny Marwayana, Graduate, UCLA

2016
|
2020

McKenzie Koch, Undergraduate, UCLA

2016
|
2019

Markus Min, Undergraduate, UCLA

2016
|
2019

Keira Monuki, Undergraduate, UCLA

2016
|
2018

Bridget Foy, Undergraduate, UCLA

2016
|
2017

Jack Garrett, Undergraduate, UCLA

2015
|
2018

Taylor Ely, Undergraduate, UCLA



TEACHING EXPERIENCE

2017
|
2019

Teaching Assistant for Introduction to Marine Science
Fall Quarters 2017, 2018 & 2019.

2015

TA for Stanford's Jasper Ridge Biological Preserve
Winter and Spring Quarters 2015.

2013
|
2014

Teaching Assistant for Photographing Nature
Winter Quarters 2013 & 2014.

2012

TA for The Global Warming Paradox
Fall Quarter 2012.



JOURNAL REVIEWS

Nature Ecology and Evolution, Nature Communications, Scientific Reports, Molecular Ecology Resources, Methods in Ecology and Evolution, PeerJ, Plos One, ICES Journal of Marine Science Environmental Science and Technology Letters, Frontiers in Ecology and the Environment, Journal of Systematics and Evolution, Environmental DNA, Coral Reefs, Copeia



PROPOSAL REVIEWS

NOAA Ocean Acidification Program, NOAA Sea Grant, National Geographic Young Explorers Program

SKILLS

Certifications

UCLA IoES Leadership in Sustainability Certificate Program. PADI rescue diver.
Diver Alert Network (DAN) insured.

Areas of Expertise

Marine ecology, eDNA, molecular ecology, bioinformatics, biological
oceanography